



AI Playbook and Toolkit for Public Health Departments

Research, Evaluation, Quality Improvement, and Public Health Practice Boundaries in AI Projects

POL 220

This module helps public health staff distinguish between AI-supported public health practice, program evaluation, quality improvement, research, and human subjects research. AI projects often blur these boundaries because they may use operational data, test new analytic methods, produce generalizable insights, or affect services and decisions. Correct classification matters because it determines what review, consent, documentation, data sharing, publication, and oversight requirements may apply. This module does not provide legal or IRB advice. Agencies should consult their legal counsel, privacy officers, IRB or research review officials, Tribal review bodies when applicable, data governance staff, and leadership. The purpose of the module is to help teams ask the right questions before AI projects begin.

Level	Intermediate
Primary track	Policy
Appears in tracks	policy

Learning Objectives

- Distinguish among public health practice, program evaluation, quality improvement, research, and human subjects research in AI project planning.
- Identify project features that may trigger IRB, legal, privacy, Tribal, or data governance review.
- Assess secondary use, derived artifact, and publication risks in AI projects.
- Create a review pathway screening memo for a proposed AI project.
- Explain why classification should occur before data extraction, modeling, or platform testing begins.

Module Overview

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Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, cybersecurity, Tribal consultation, labor relations, human resources, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

Why This Topic Matters for Public Health AI

Public health agencies routinely collect and use data for surveillance, operations, evaluation, emergency response, and service improvement. AI projects may start as internal improvement efforts but later generate methods, models, publications, or tools that look like research. If teams do not identify the boundary early, they may miss required review or fail to communicate clearly with partners and communities.

AI also raises new questions about secondary data use, model development, automated analysis, external platforms, reproducibility, and reuse. A project may not involve direct interaction with people but may still use identifiable health data or community-level data in ways that require oversight. Teams should classify the activity before data are shared, linked, uploaded, modeled, or reused.

Public Health Example

A health department wants to use AI to summarize years of case investigation notes to identify workflow bottlenecks. The initial purpose is internal quality improvement, but the team also wants to publish a paper describing the AI method and compare results across jurisdictions. The project uses identifiable historical records during model testing.

Before work begins, the team consults privacy, legal, data governance, and IRB officials. They clarify whether the activity is quality improvement, research, or both; whether identifiable data can be used; whether data must remain in an approved environment; whether publication changes the review pathway; and whether partners must approve reuse. The project plan is revised to separate internal improvement from any research component.

Technical, Programmatic, and Operational Deep Dive

Classification should begin with intent, data, methods, risk, and dissemination. Teams should ask why the project is being done, who benefits, whether the activity is required for public health operations, whether the methods are experimental, whether individuals or communities could be affected, whether identifiable or sensitive data are used, and whether results will be generalized or published. These questions help reviewers determine the appropriate pathway.

AI model development can add complexity because data may be transformed into features, embeddings, prompts, outputs, derived datasets, or trained models. These artifacts may retain information from the original data or create new risks. Reviewers should consider not only raw data but also intermediate and final AI artifacts.

Teams should also consider partner and community expectations. Data collected for public health practice may not be appropriate for AI research or product development without additional review or agreement. This is especially important when data involve Tribal Nations, small communities, vulnerable populations, community partners, or data sharing agreements with specific use limits.

Risks, Failure Modes, and Guardrails

A common failure mode is assuming that all internal AI work is quality improvement and therefore needs no additional review. Some internal projects may involve identifiable data, experimental methods, high-risk outputs, or plans for publication. Guardrails include early classification review, documentation of purpose, and consultation with IRB or research officials when uncertain.

Another risk is secondary use beyond the original data purpose. AI teams may want to reuse operational data to train, test, or validate models in ways that were not anticipated. Guardrails include reviewing data use agreements, consent or notice terms, legal authority, partner expectations, and community implications before reuse.

A third risk is failing to manage derived AI artifacts. Prompts, outputs, embeddings, labels, trained models, and evaluation datasets may contain or reveal sensitive information. Guardrails include approved environments, retention rules, access controls, deletion procedures, documentation, and review before sharing or publication.

Application to AI-Supported Workflows

AI project intake should include a research-versus-practice screening question. If the project involves identifiable data, novel methods, publication intent, multi-jurisdiction comparison, human subjects, Tribal data, or partner data reuse, the project should be routed for additional review before work begins. Screening should occur before data extraction or platform testing.

Project documentation should state the classification decision and the basis for that decision. It should also state what approvals were obtained, what data may be used, what outputs may be shared, and what changes would trigger re-review. This documentation supports accountability and helps prevent scope creep.

Reflection Questions

What is the primary intent of the AI project: operations, evaluation, quality improvement, research, or more than one?

Will the project use identifiable data, partner data, Tribal data, or data with use restrictions?

Could publication, model sharing, or reuse change the review pathway?

Practical Exercise

Create a review pathway screening memo for a fictional AI project. Choose a use case such as summarizing case investigation notes, evaluating a chatbot pilot, building a risk model, analyzing public comments, or testing AI

transcription for interviews. Use fictional or generalized information only.

Your memo should describe the project purpose, data involved, whether identifiable information is used, whether human subjects may be involved, whether results may be published, whether partner or Tribal data are involved, what review bodies should be consulted, and what changes would trigger re-review.

Example: A program wants to use AI to analyze de-identified training evaluations and publish lessons learned. The memo notes that the activity may be program evaluation, but publication intent and free-text comments require review. It recommends privacy review, IRB determination, removal of identifiers, and documentation that AI outputs will not be treated as participant-level findings.

Expected Artifact or Evidence

AI Project Review Pathway Screening Memo

List of review bodies and re-review triggers

Knowledge Check

- 1. What is the most important first step after completing the Research, Evaluation, Quality Improvement, and Public Health Practice Boundaries in AI Projects module?
- 2. Which practice best supports responsible AI use in public health?
- 3. Which statement best reflects the risk or guardrail for this module: A common failure mode is assuming that all internal AI work is quality improvement and therefore needs no additional review. Some internal projects may involve identifiable data, experimental methods, high-risk outputs, or plans for publication. Guardrails include early classification review, documentation of purpose, and consultation with IRB or research officials when uncertain.
- Another risk is secondary use beyond the original data purpose. AI teams may want to reuse operational data to train, test, or validate models in ways that were not anticipated. Guardrails include reviewing data use agreements, consent or notice terms, legal authority, partner expectations, and community implications before reuse.
- A third risk is failing to manage derived AI artifacts. Prompts, outputs, embeddings, labels, trained models, and evaluation datasets may contain or reveal sensitive information. Guardrails include approved environments, retention rules, access controls, deletion procedures, documentation, and review before sharing or publication.
- 4. When should a staff member escalate an AI-related concern?
- 5. What counts as stronger completion evidence for a training module?

References and Resources

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